

ANALYSTLAB AFRICA

Week 3: SQL & Data Querying

Dataset: Sample Sales Data

1. DATABASE SETUP

This dataset came from Kaggle. It has 2,823 rows showing sales orders from different countries, years, and product types.

2. CORE QUERIES

I used SELECT, WHERE, and ORDER BY to find shipped orders and large deals. I used GROUP BY and HAVING with SUM and AVG to find total revenue by product line, country, and year.

Findings: Classic Cars is the top product line (\$3.9M). USA is the top country (\$3.6M). 2004 was the best year (\$4.5M).

3. ADVANCED QUERIES

- CASE WHEN: I split revenue into Q1, Q2, Q3, Q4 columns for each year. Q4 is always the strongest quarter.
- Subqueries: I found customers who spent more than the average. I also found products priced above their group's average.
- Window Functions: I used RANK() to rank customers by revenue in each country and ROW_NUMBER() to find the top 3 products in each product line.

4. KEY FINDINGS

- Top customer: Euro Shopping Channel from Spain (\$912,294)
- USA has the most large deals (64), followed by France and Spain
- Loyal customers (6+ orders) spend 8 times more on average than occasional buyers
- Revenue grew 32% from 2003 to 2004

5. QUERY OPTIMISATION

I added indexes on STATUS, YEAR_ID, PRODUCTLINE, CUSTOMERNAME and COUNTRY since these columns were used the most in my queries.

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